OOD + Alignment → Category-Shift Adaptation

A cheap `partial_fit` recipe for new-category cold-start

AS-DTA / CFPerm

Abstract

One-liner. Quantify category shift as OOD, then adapt a CE head with three cheap weights via `partial_fit`: feature (VIMP), sample (membership), alignment (hop/prototype cosine). Soft$^{\gamma=0.5}$ tapers empowerment by 200 samples/category. The polished online recipe is **recommended**: `both+align` while the new category is scarce, then `both`. On text multiclass (FULL LIVE holdout), recommended lifts early acc (0.456 → **0.481** at 20/cat) and finishes at +**0.014** overall vs uniform.

1 Recipe (what to ship)

Step	Object	Role
1. Quantify	RF-domain VIMP v, membership m	Where LIVE differs from REF
2. Align	align $= \sigma(\beta(\cos_{\text{nov}} - \cos_{\text{known}}))$	High alignment with novel direction → up-weight
3. Adapt	feature / sample / align → `sample_weight`	CE `partial_fit` only
4. Gate	soft$^{\gamma=0.5}$, $N_{\text{cap}} = 200$	Empower, then taper — no EWMA

Recommended online policy.

$$\texttt{recommended} = \begin{cases} \texttt{both+align} & \text{if cold-start frac } \geq 0.75, \\ \texttt{both} & \text{otherwise.} \end{cases}$$

Use alignment when the new category is scarce; drop it once most novel mass is past the cap. Still one linear head.

2 Evidence (FULL LIVE holdout)

	uniform	both	both+align	**recommended**
acc @ b=20/cat	0.456	0.475	**0.481**	**0.481**
endpoint acc	0.636	0.647	0.650	**0.650**
endpoint Δacc	—	+0.011	+0.013	+**0.014**

Cost. One shallow RF + a few CE `partial_fit` epochs. No new backbone, no EWMA schedule.

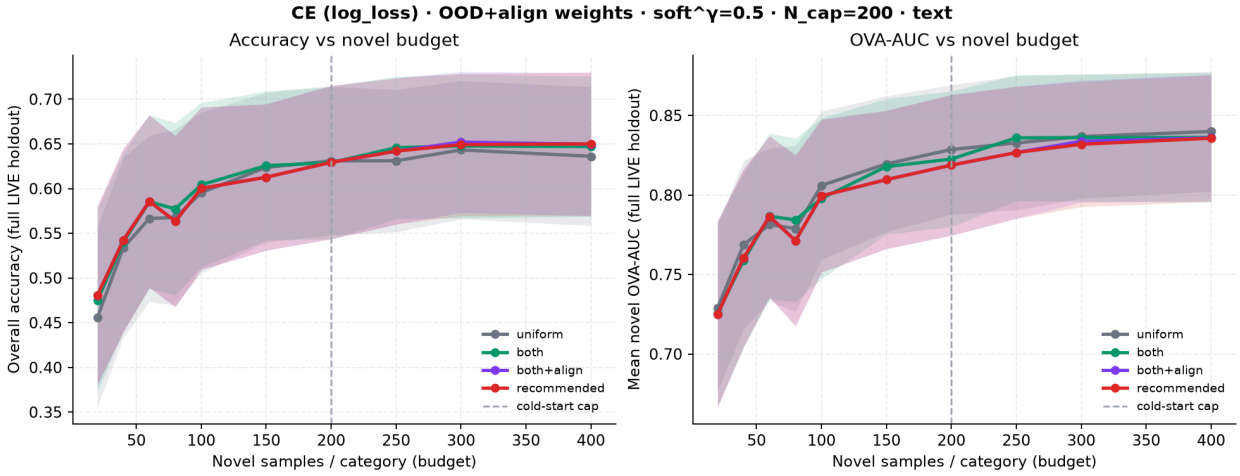


Figure 1: Polished comparison (mean±SEM): uniform, both, both+align, recommended. Dashed line: $N_{\text{cap}} = 200$. Recommended tracks both+align early and stays on top late.

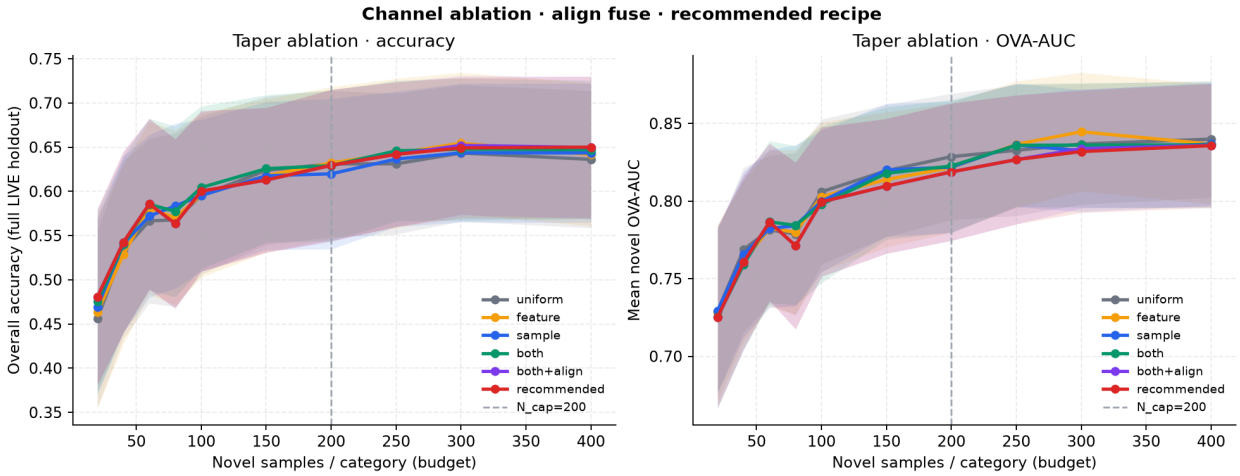


Figure 2: Channel ablation. Feature / sample / both / align / recommended. Recommended is the ship default.

3 Takeaway

1. Category shift = OOD; quantify once, adapt with weights.
2. Alignment is the same "high align $\rightarrow$ upweight" family as hop-cosine IW — plug into `sample_weight`.
3. Ship **recommended**: align while scarce, then both.